

Part II.

Results

In this part we are going to describe the computational and analytical methods developed. The main goal is to improve the accuracy of different MHD simulation setups without a significant tradeoff in computational cost. This was done in 2 different approaches:

- **Post-hoc:** In this approach a superresolution model is applied to the final state of a low resolution simulation in order to try to recover statistics such as power spectra and meaningful structures.
- **Ad-hoc:** In this approach a neural network is intertwined with the simulator, learning how to steer it in the right direction in order to get close to a high resolution simulation baseline downscaled to the low resolution grid.

4. Post-hoc approach: Super resolution

Super resolution models aim to transform a low resolution input, typically 2 dimensional 3 channel images captured by cameras, to a higher resolution image that reconstructs the lost information in the downsampling operation. These downsampling operations are typically caused by technical constraints. In the case of traditional cameras their sensors are made of squared light receptors, the density of which determines the final resolution of the image.

In our case the technical constraint is by mere experiment design as the actual resolution of the simulation is chosen by the experimenter at the early design stages and it can be compromised by factors such as time, memory or energy. Thus low resolution simulations are preferred costwise. From a numerical analysis point of view, given a stable and consistent numerical method, the global error will converge at the same rate as the truncation error, the error introduced in a single simulation step. Thus if the truncation

error is reduced as the temporal or grid resolution ($\Delta t, \Delta x$) are reduced ($\Delta t \rightarrow 0, \Delta x \rightarrow 0$), so will do the global error⁹ [LeVeque, 2002, Chapter 8.2]. Therefore higher resolution simulations will be more accurate to the physical model, governed by the PDE, than lower resolution simulations.

Super resolution is an ill defined inverse problem due to the non affinity of the down-sampling operation, this is, an infinite amount of high resolution states could correspond to its low resolution pair. The solution to this ill defined problem is defining priors over the high resolution states, in the case of machine learning this means that our model has to understand the dynamics that result in the HR states and therefore understand the underlying physical model. Therefore we develop neural operator models with physically informed strategies to superresolve turbulent slices. Our approach is a proof of concept for neural operators within the context of turbulence superresolution and it is not intended to be a valid methodology but rather a study of feasibility.

4.1. Dataset

The dataset used consists of 500 pairs of high and low resolution turbulence simulations split into training (400 pairs) and validation (100 pairs) sets, each state of the simulation consists of a 5 channel (pressure, 3 velocity fields and density) 3 dimensional state. Each initial velocity state is obtained by building a random phase Gaussian field with a predefined power spectrum shape in Fourier space (following Kolmogorov's law) between two cutoff wave numbers, to then apply the inverse Fourier transform to it. The velocity field is then scaled to correspond to a predefined mean velocity, in our case this corresponds to 50 km/h.

The pressure and density fields are initialized as constant and uniform with values $\rho_0 = 2m_p \text{ cm}^{-3}$ and $p_0 = (3 \times 10^4 \text{ K}) k_B \text{ cm}^{-3}$, where m_p is the proton mass and k_B Boltzmann's constant. The end time is 10000 years and the physical box is 3 parsecs long in each direction. The integration was done on the finite volume mode of the simulation with a CFL value of 0.4. The high resolution that will be used as the targets for the loss

⁹ $\Delta t, \Delta x$ are related through the Courant–Friedrichs–Lewy condition as $C = u\Delta t/\Delta x$ which for many discretizations is a needed condition to ensure stability

calculation is 128^3 , whereas the low resolution is 32^3 . To store and load the states we use the python library h5py which allows for a convenient and organized storage.

Using the speed of sound formula for an ideal gas $c_s = \sqrt{\gamma P / \rho}$ and $\gamma = 5/3$ we get a value of $c_s = 14.4$ km/s, given the mean velocity we obtain a Mach number of $M = U_{rms}/c_s \approx 3.5$, which corresponds to a highly supersonic regime.

4.2. Considered superresolution models: Fourier Neural Operator and Enhanced Deep Residual Network

Two models were considered and compared, a Fourier neural operator (FNO) similar to the one in section 2.6.3 adapted to superresolution in [Yang et al., 2023] and a ResNet based superresolution model based on [Lim et al., 2017]. We introduce an extra term to the FNO which shifts the Fourier modes acted on, hence we call this model Shifted-FNO, or SFNO.

4.2.1. Shifted Fourier Neural Operator

Starting from the SFNO, the operator G_θ acting on a state $s \in \mathbb{R}^{d_{LR}^3}$ is built as

$$G_\theta(s) := \mathcal{F}_\theta(T^{-1}(f_\theta(s))) \in \mathbb{R}^{d_{HR}^3}, \quad (46)$$

where f_θ is the initial lifting operator, here implemented as a ResNet, T^{-1} is a discretization inversion operator, here implemented as a trilinear interpolation, and $\mathcal{F}_\theta$ is functional operator, here implemented as an FNO following the original paper. The discretization inversion operator is chosen as an interpolation with the assumption that the FNO is complex enough to learn the proper mapping. Going into the detail of each of the parts:

- **Lifting network f_θ :** modelled with a set of standard ResNet blocks with 3x3x3 convolutions, with a ReLu as activation function. Transforms the input to a higher dimensionality space given by the number of hidden channels C_{hidden} , thus $f_\theta(s) \in \mathbb{R}^{C_{hidden} \times d_{LR}^3}$

- **Discretization inversion operator** T^{-1} : interpolates $f_\theta(s)$ in latent space to the target resolution with a variable scaling factor, $T^{-1}(f_\theta(s)) \in \mathbb{R}^{C_{hidden} \times d_{HR}^3}$. Thus, as the interpolation is done on arbitrary target grids without being fixed to a integer scale factor as in the EDSR architecture, the architecture is resolution agnostic and we can say we are talking about a neural operator.
- **Functional operator** $\mathcal{F}_\theta$: modelled as an FNO, learns the mapping from the interpolated high dimensional $T^{-1}(f_\theta(s))$ to the intended dimensionality (3 in our case). It holds the majority of the responsibility in the upsampling operation and as such it has to be expressive enough. The FNO is defined by a series of Spectral Blocks which apply both a spectral convolution and a 1x1x1 kernel convolution which get joined at the end.
- **Projection layer**: a final layer maps back to the input channels, this is done by a 3x3x3 convolution together with a softmax constraint layer from [Harder et al., 2024] to ensure mass and energy conservation. It also allows for a ReLu pass on the density and pressure channel to preserve positivity.
- **Final (optional) skip layer**: an optional skip layer with the initial input upsampled by an interpolation can be applied, delegating the FNO to compute not the whole image but the residual distinctions with the interpolation.

We introduce in this work an extra parameter to the spectral convolution, which controls the region of the Fourier space the learned parameters act on. Typically in a spectral convolution this is not controlled and the parameters act from the 0th wave number to a predefined cutoff value k_{max} , by introducing an extra parameter called k_0 or shifting modes, we modify the region of the Fourier space acted on to $[k_0, k_{max} + k_0]$ and $[-k_{max} - k_0, -k_0]$. The motivation for the introduction of such a parameter lies in the unresolvedness of the finer details in Fourier space, rather than the larger scales. By shifting the action of the network to this finer scales we aim for the network to be able to predict the smaller scales lost in the low resolution simulation. At the same time the coarser scales are preserved by the piecewise section of the Spectral blocks. As we are shifting the region of application of the FNO layer, we call this architecture Shifted-FNO

(SFNO).

4.2.2. Enhanced Deep Residual Network

The CNN model used is based on the Enhanced Deep Residual Network (EDSR) model from [Lim et al., 2017]. Here the network was adapted to allow for 3 dimensional inputs whereas in the original implementation it was for 2 dimensional inputs. The network consists of 3 distinct blocks, the lifting block, feature extraction block and upsampling block, applied in such order. Going on detail of each of the blocks:

- **Lifting block:** A single 3x3x3 convolution lifts the 5 channel 3D input to a state with n hidden channels.
- **Feature extraction:** A stack of residual blocks, each of the residual blocks consists of two 3x3x3 convolutions with a ReLu between them and an identity skip connection. After applying this block another skip connection is applied with the output of the lifting block.
- **Upsampling:** Done by the application of n PixelShuffle layers [Shi et al., 2016], where n is given by the scale factor $s = 2^n$. A PixelShuffle application consists of a 3 dimensional convolution that multiplies the number of channels by 8, and then a pixel shuffle operation which rearranges the expanded channels into 2x2x2 spatial blocks, effectively doubling the resolution. Finally a 3x3x3 convolution brings down the number of channels to the desired one.

Comparing both architectures, the EDSR has 2 clear shortcomings against the FNO. Firstly the EDSR is forced into a scaling factor from the definition, whereas the FNO allows for arbitrary scale factors. Secondly the EDSR is a pure local convolution architecture, whose effective field of interaction is given by the number of residual blocks (2 per residual block application), on the other hand the FNO has a global interaction field due to working in Fourier space which can be exploited for turbulent fields with long correlations.

4.3. Benchmarks: SFNO and EDSR models

After running an Optuna optimisation we find the best parameters for the models considered which are given in table 3. We observe how the best shifting modes found correspond to 8, favouring high wave numbers and ignoring the low wave numbers which are captured by the residual connections. The metrics table for both models and interpolation as a benchmark can be seen in Table 4; the MSE per channel, on the energy spectra and on the vorticity are presented. We observe how the SFNO outperforms the EDSR slightly. The models were trained for 500 epochs and 250 epochs in the case of SFNO and EDSR respectively, each epoch corresponding to computing the whole training dataset on shuffled batches of 64. All the training hyperparameters are presented in appendix A.

Table 3: Model hyperparameters for the SFNO and EDSR architectures.

Parameter	SFNO	EDSR
Input channels	5	5
Feature channels	32	64
Residual blocks	3	16
Operator blocks	2	–
Fourier modes	16	–
Shifting modes	8	–
Kernel size	–	3
Skip connection	True	True
Number of parameters	33.5 mill	5.5 mill

The loss evolution curves of both the SFNO and EDSR model can be seen in Figure 4.2. The SFNO does not show a substantial improvement to the performance of the EDSR, one factor that may contribute to the similar performance is the similarity between the training and evaluation data. Since the data was generated with the same parameters and just changed the initial state velocity distribution, the performance on the evaluation data does not represent the ability of the model to generalize to other regimes, therefore, the potential improvement of the SFNO may not be reflected in this evaluation. We observe a slight overfitting for the SFNO, this can be explained due to the high amount of parameters compared with the EDSR. In order to observe a larger difference between the two models the dataset should span a larger range of parameters and the validation

Table 4: MSE metrics for $\times 4$ upsampling.

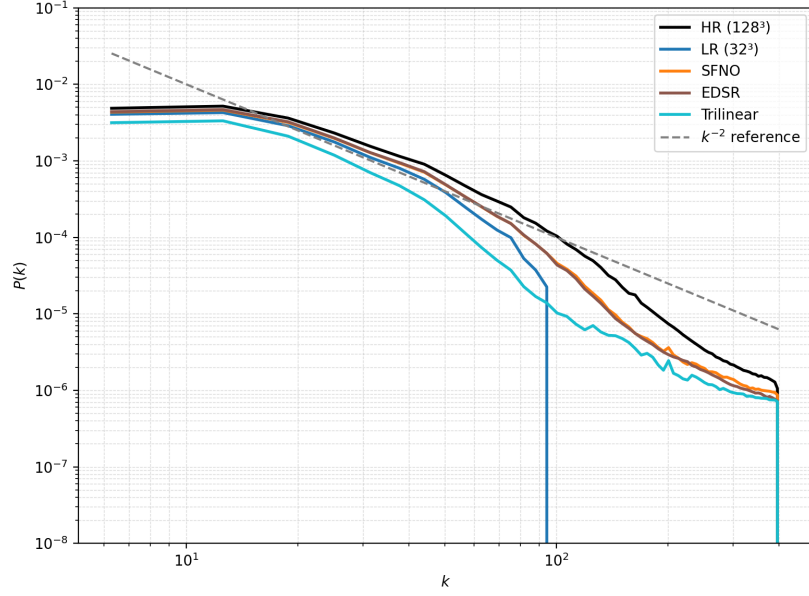
Metric	EDSR	SFNO	Trilinear
MSE	0.008 47	0.00808	0.037 16
MSE Pressure	0.000 31	0.00028	0.001 63
MSE Density	0.038 15	0.03643	0.162 08
MSE v_x	0.001 28	0.00121	0.007 33
MSE v_y	0.001 30	0.00124	0.007 38
MSE v_z	0.001 29	0.00124	0.007 40
MSE $\ v\ $	0.001 48	0.00141	0.008 16
MSE Vorticity	0.001 07	0.00100	0.002 84
MSE Energy Spectra	0.066 39	0.04987	0.217 37

data contain simulations generated with unseen parameters. Nevertheless we still see improvement from the SFNO to the EDSR. A snapshot and energy spectra comparison for both models and interpolation can be seen in Figure 4.1.

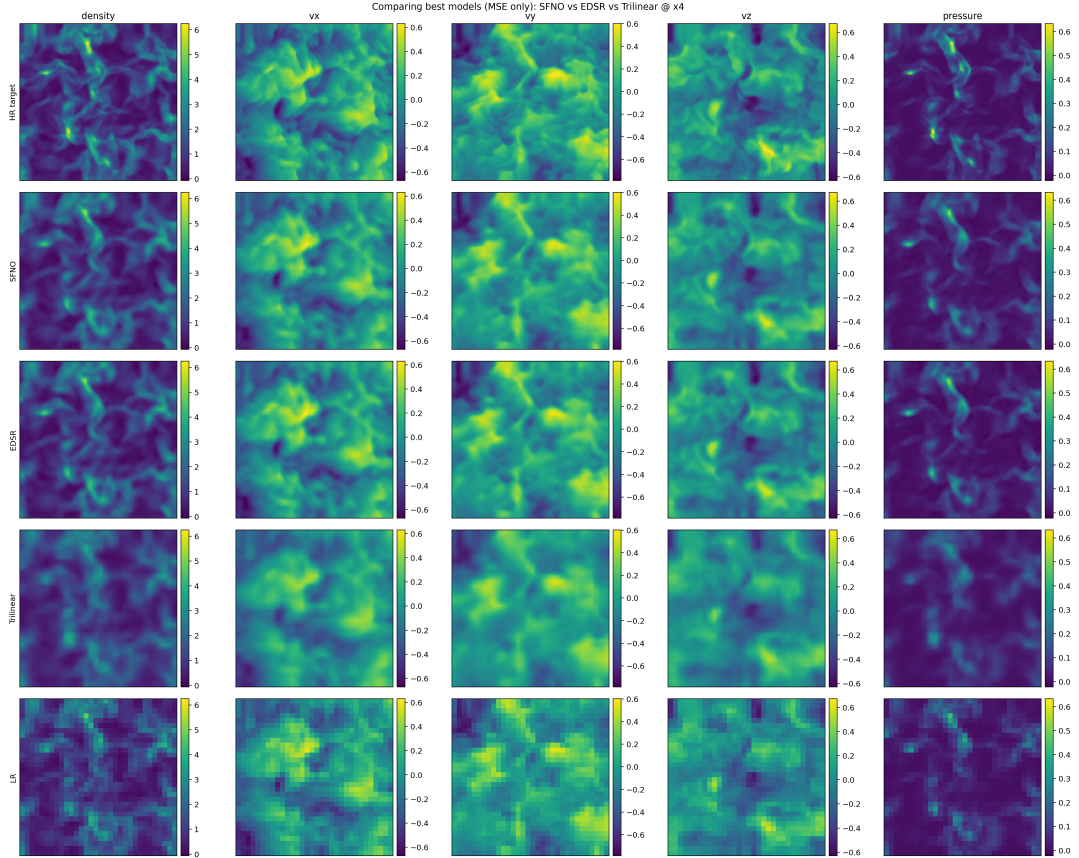
4.4. Metrics study: L1 and spectral loss

In the previous section the models were trained with an MSE loss, although suitable for most machine learning problems in the case of superresolution MSE averages out the details and doesn't allow the model to learn sharper edges. Thus we consider two different metrics to train on: the L1 loss and a spectral MSE from the velocity spectra. This last one is physically motivated from section 3.3.3 and the Kolmogorov spectra. The L1 loss is motivated by [Zhao et al., 2018] in which it is claimed that in order to achieve perceptually better and sharper images L1 losses are better posed than L2 based losses.

We test these additional losses by adding each as an extra term to the MSE with a weight that represents 10% of the MSE. 3 models were trained, 2 with MSE and each of the extra losses (L1 and Spectral) and one with the 3 losses. The results can be seen in table 5. The combination of MSE and L1 seems to yield the best MSE value by a small margin whereas the MSE and spectral loss combination yields similar values to the other models while improving the performance on the spectral MSE. This can be

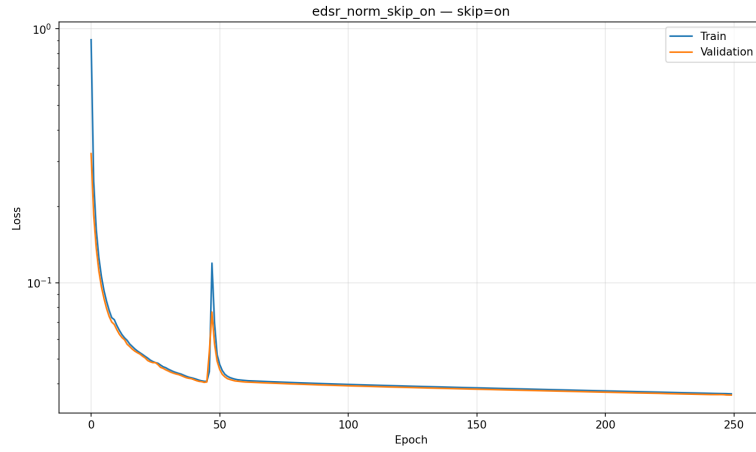


(a) Spectra comparison of a high resolution end state and low state and superresolved states from interpolation, an EDSR model and an SFNO model.

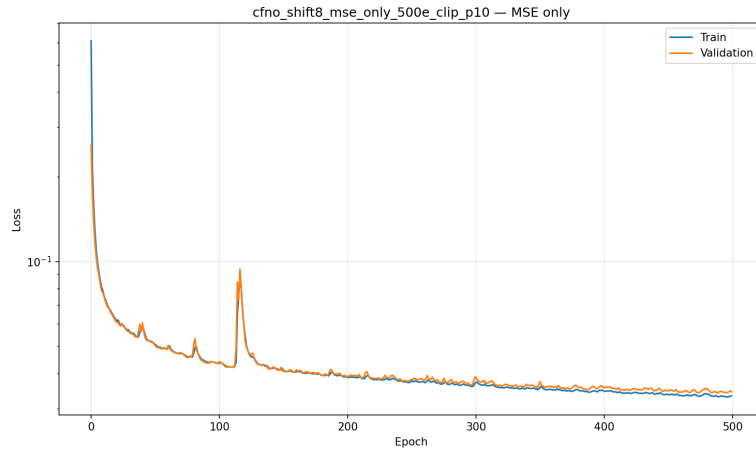


(b) Snapshot slice of a turbulent superresolved state with interpolation, an EDSR model and an SFNO model.

Figure 4.1: Spectra and snapshot comparison of SFNO and EDSR superresolution models.



(a) Loss evolution of EDSR model trained with parameters from table 3



(b) Loss evolution of FNO model trained with parameters from table 3

Figure 4.2: Loss evolution of both FNO and EDSR models trained for superresolution of turbulent states.

Table 5: Comparison of loss functions for the SNO model with $\times 4$ upsampling. Lower is better.

Metric	MSE	MSE + Spectral	MSE + L1	MSE + Spectral + L1
MSE	0.008 08	0.008 20	0.00799	0.008 50
MSE Pressure	0.000 28	0.000 29	0.00028	0.000 30
MSE Density	0.036 43	0.036 92	0.03609	0.038 27
MSE v_x	0.001 21	0.001 26	0.00118	0.001 32
MSE v_y	0.001 24	0.001 25	0.00120	0.001 31
MSE v_z	0.001 24	0.001 28	0.00120	0.001 31
MSE $\ v\ $	0.001 41	0.001 41	0.00136	0.001 47
MSE Vorticity	0.001 00	0.001 00	0.00099	0.001 03
MSE Spectra	0.049 87	0.03274	0.046 71	0.040 42

observed in Figure 4.3, although the effect is rather light. In the following we train on the MSE + Spectral loss as for a little trade off in MSE the velocity spectra improves considerably which is one of the most important statistics we would like to recover.

4.5. An expansion to the SFNO model: U-Net SFNO

We test another architecture from [Wen et al., 2022] in which an extra path to the spectral block is added, this time in the shape of a U-Net, hence the name UFNO. In the original proposal both spectral blocks and U-Net spectral blocks were used, in our case as our model consists of 2 spectral blocks we substitute both. A scheme for a UFNO can be seen in Figure 4.4, we illustrate the shifting of the modes in the spectral blocks, resulting in the architecture we refer to as U-SFNO.

The model is trained following the same parameters as the SFNO models from the previous section. The loss used is MSE + Spectral. The resulting metrics on the evaluation dataset can be seen in table 6. The U-SFNO model outperforms the two other models in every metric, when comparing a slice of a snapshot and its energy spectra, shown in figures 4.5 and 4.6. Despite the improved performance of the U-SFNO, all three models considered preserve the overall structure of the flow while struggling to accurately recover the sharpest peaks and fine scale features.

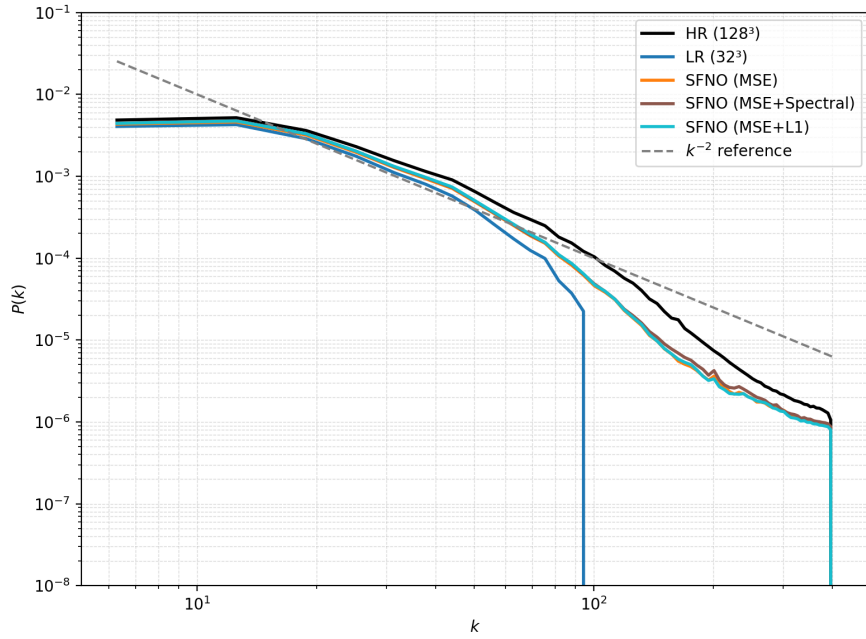


Figure 4.3: Spectra comparison of SFNO models trained with different losses, MSE was used as basis and 2 extra terms are studied. An spectra based loss to physically inform the model and an L1 loss to recover sharper edges.

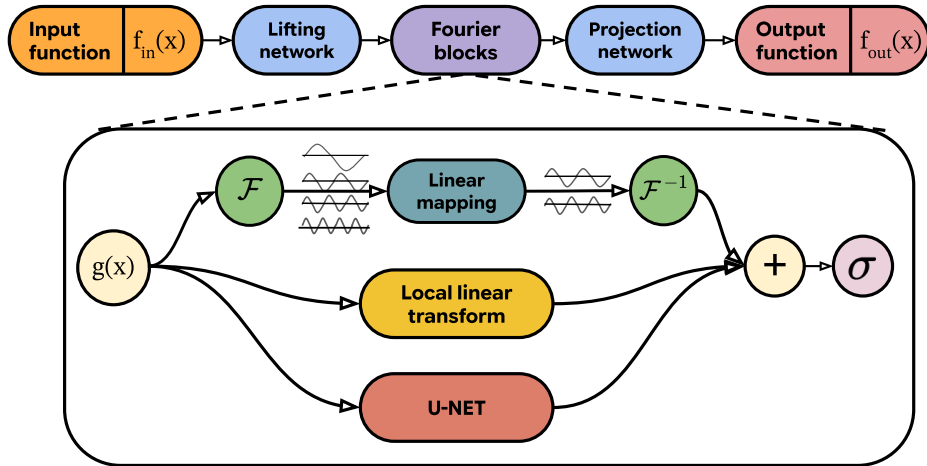
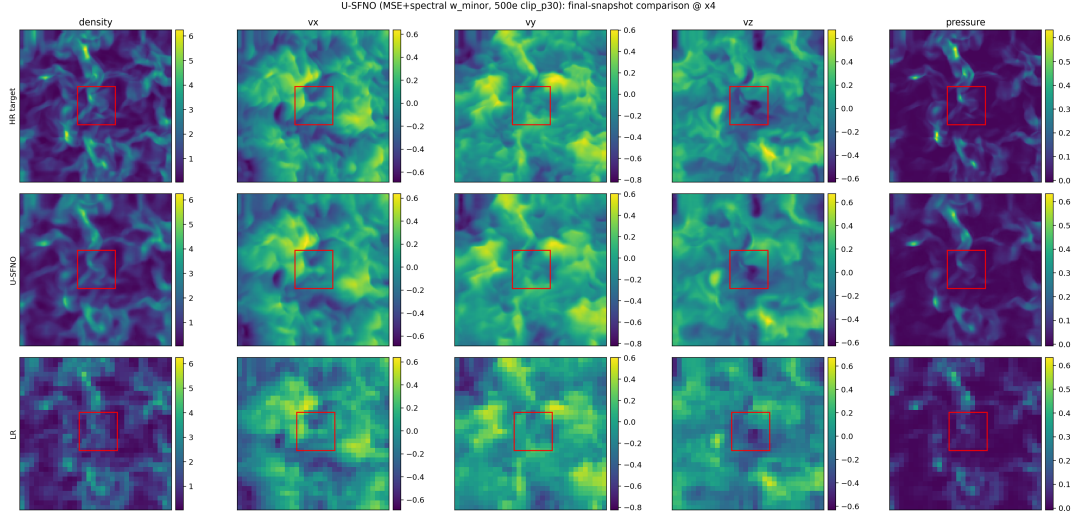
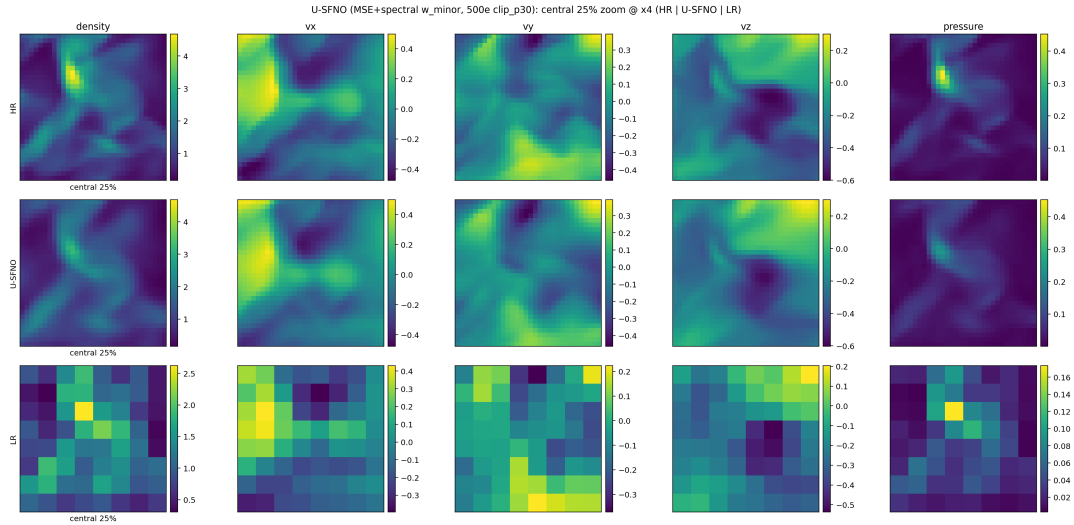


Figure 4.4: U-SFNO full architecture. Similar to a FNO architecture, the state is lifted to a higher dimensional space, followed by the application of the spectral blocks and later transformed back to the original dimensionality. The difference lies in the spectral blocks, which now incorporate an extra path encoded by a U-Net.



(a) Turbulent snapshot slice of a high resolution, low resolution and superresolved states with the U-SFNO model. The red rectangle represents the zoomed in region shown in the figure below.



(b) Zoomed in slice of a turbulent snapshot. Presented here are the high resolution simulation, superresolved and low resolution input of the SR model, in this case a U-SFNO.

Figure 4.5: U-SFNO application on a slice of a turbulent snapshot. In both plots HR (top row), U-SFNO augmented (middle row) and LR (bottom row) are presented.

Table 6: Comparison of spectral loss for operator networks with $\times 4$ upsampling. Lower is better.

Metric	SFNO MSE	SFNO MSE + Spectral	U-SFNO MSE + Spectral
MSE	0.008 08	0.008 20	0.00613
MSE Pressure	0.000 28	0.000 29	0.00020
MSE Density	0.036 43	0.036 92	0.02767
MSE v_x	0.001 21	0.001 26	0.00090
MSE v_y	0.001 24	0.001 25	0.00094
MSE v_z	0.001 24	0.001 28	0.00093
MSE $\ v\ $	0.001 41	0.001 41	0.00104
MSE Vorticity	0.001 00	0.001 00	0.00074
MSE Spectra	0.049 87	0.032 74	0.02407

4.6. Discussion

Superresolution for turbulent hydrodynamical setups has been tested on a dataset covering a limited amount of parameters with succesful results. Overall the tests presented here are rather reduced and a full fledge test would need generalization results and a larger dataset. Still as an early test super resolution has shown to work under a turbulent setup for interstellar medium applications. In climate sciences super resolution driven methods, referred as downsampling, have been proven to work under the chaotic system that is the earth’s climate [Najafi et al., 2026]. More recently super-resolution methods have also been developed for turbulent setups with NO approaches [Oommen et al., 2025] and [Abid and San, 2026]. Thus as a first approach such an application to the ISM was expected to work. A full comparison with emulator based models is also needed to ensure which of the two methods gives better results on long simulation times.

Commenting on the overall performance, we’ve seen a well known problem with super-resolution setups, the lack of finer structure. Although we tried to circumvent it with the FNO and the shifted fourier modes it still appeared due to using MSE based losses. Further study and optimisations are needed to the model and training procedure. In the model aspect diffusion and transformer based models have been proven to recover finer details and sharper edges than traditional networks or neural operators. Still staying in

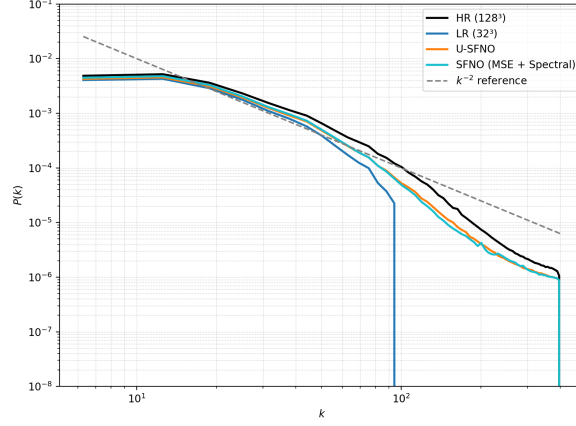


Figure 4.6: Energy spectra comparison of a high resolution and low resolution turbulent states with the U-SFNO model and the SFNO model both trained with an MSE + Spectral combined loss.

the neural operator regime, other approaches using FNOs exist like in [Qin et al., 2024] or [Abid and San, 2026] both multiscale approaches. Other techniques to be studied are the use of gradient based losses or other combinations of losses that might help recover the finer details or adversarial training strategies as in GANs.

As to the effectiveness of a the neural operator against a normal neural network architecture we can comment on the fact that such a model allows for other increments of superresolution beyond the ones it was trained on. Thus one could train a model for 4x superresolution and use it for 2x out of the box or fine tune it even more, which would allow for multiscale methodologies, although this needs further testing. It was also shown the efficiency of the shifted fourier modes which allow the model to capture finer details in fourier space which are key for turbulence.